

Rai Yashmit

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SKILLS

Languages: C++, C, Python, SQL
Tech Stack: Power BI, Excel, Docker, Git
Frameworks: TensorFlow, Scikit-Learn, Fast API, NumPy, Pandas
Core Skills: Data Structures and Algorithm, System Design, Machine Learning, Data Analysis
Soft Skills: Strategic Thinking, Adaptability, Rapid Skill Acquisition, Leadership

PROJECTS

Denoising Diffusion Probabilistic Model (DDPM) for Image Generation on MNIST Dec' 25

- Engineered a **16.37M-parameter conditional DDPM** in TensorFlow, trained on **60k MNIST images**, generating 28x28 digits with a cosine schedule and Classifier-Free Guidance.
- Reduced reconstruction loss by **~10%** and maintained a train-validation loss gap **<0.001**, indicating strong **generalization**.
- Built a "Sketch-to-Digit" pipeline using an **870K-parameter CNN** classifier with higher than **99% accuracy**.

Ollama-Powered RAG Assistant for Particle Physics | Python, Ollama, Streamlit Nov' 25

- Developed a **Retrieval-Augmented Generation (RAG) pipeline** using Ollama, Llama 3.2, and Streamlit to deliver context-aware answers for particle physics queries through **fully local LLM inference**.
- Implemented an embedding and retrieval pipeline using **nomic-embed-text**, and **in-memory vector database with cosine similarity** for semantic search and efficient document retrieval.
- Orchestrated LLM inference, prompt engineering, and context injection by retrieving **top-ranked knowledge chunks** before response generation, **improving factual accuracy and reducing hallucinations**.

TRAINING

Library Management System | Python, SQLite, FastAPI Aug' 25

- Architected a RESTful backend system using FastAPI, handling 20+ endpoints for user, inventory, and transaction management.
- Integrated a user preference engine** supporting multi-attribute subscriptions and implemented an **Observer Pattern-based notification system** for real-time book availability alerts.
- Designed a normalized relational schema (3NF) supporting **5+ entities and complex relationships (1:N, M:N)**.

CERTIFICATES AND CERTIFICATIONS

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|---------------------------------------|---------|
| • Data Science Oracle | Oct' 25 |
| • Data Structures and Algorithm LPU | Jul' 25 |
| • Elite in Cloud Computing NPTEL | May' 25 |

EDUCATION

Lovely Professional University	Phagwara, Punjab
Bachelors of Technology in Computer Science and Engineering	Aug' 23 - Present
CGPA – 8.87	